Major Project Title

submitted in partial fulfillment of the requirement
for the award of the Degree of

Bachelor of Technology
in
Information Technology

by

Student1 Name and Surname
Student 2 Name and Surname
Student 3 Name and Surname

under the guidance of

Guide Name
Co-Guide Name



Department of Information Technology
Bharatiya Vidya Bhavan's
Sardar Patel Institute of Technology
(Autonomous Institute Affiliated to University of Mumbai)
Munshi Nagar, Andheri-West, Mumbai-400058
University of Mumbai
2020-2021

Approval Certificate

This is to certify that the Project entitled "Major Project Title" by Student 1, Student 2 and Student 3 is approved for the partial fulfillment of Major Project - I for the Final Year Project towards obtaining the Degree of Bachelor of Technology in Information Technology from University of Mumbai.

Project Guide

(signature)

Name:

Date:

External Examiner

(signature)

Name:

Date:

Head of Department

Principal

Seal of the Institute

Declaration

I wish to state that the work embodied in this work titled "project title" forms my own contribution to the work carried out under the guidance of Guide Name at the Sardar Patel Institute of Technology. I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission.

Candidate 1 Name	**Candidate 2 Name**	**Candidate 3 Name**
Candidate 1 UID	**Candidate 2 UID**	**Candidate 3 UID**

Abstract

Place your abstract here

List of Figures

Add your list of figures here

List of Tables

Add your list of tables here

Contents

1 Introduction **7**
1.1 Problem Statement 7
1.2 Objectives 7
1.3 Scope 7
1.4 Technologies Used 7

2 Literature Survey **7**
2.1 Literature Survey 1 7
2.2 Literature Survey 2 7

3 Analysis **7**
3.1 Analysis diagram 1 7
3.2 Analysis diagram 2 7
3.3 Analysis diagram 3 7

4 Design and Methodology **8**
4.1 Design and Methodology 1 8
4.2 Design and Methodology 2 8
4.3 Design and Methodology 3 8

5 Results and Discussion **8**
5.1 Results and Discussion 1 8
5.2 Results and Discussion 2 8

6 Conclusion and Further Work **8**

7 Research Paper **9**

8 Plagiarism Report **9**

1 Introduction

Note: Use subsections and subsubsections wherever required throughout the report
Introduce your topic here

1.1 Problem Statement

1.2 Objectives

1.3 Scope

1.4 Technologies Used

use subsections evrywhere Assumptions, constraints

2 Literature Survey

Note: Use subsections and subsubsections wherever required throughout the report
Must contain a detailed study of minimum 12 research papers on the project topic and gap finding
Use table captions to name tables at the top and use figure titles below figures wherever applicable

2.1 Literature Survey 1

2.2 Literature Survey 2

...
use page breaks after each section

3 Analysis

Note: Use subsections and subsubsections wherever required throughout the report
Use table captions to name tables at the top and use figure titles below figures wherever applicable Architecture/DFDs/class diagram/use-case diagram/sequence diagram/ flowchart/flow-diagram(others-as suggested by your guide)

3.1 Analysis diagram 1

3.2 Analysis diagram 2

3.3 Analysis diagram 3

.....
Use page breaks after each section

4 Design and Methodology

Note: Use subsections and subsubsections wherever required throughout the report
Use table captions to name tables at the top and use figure titles below figures wherever applicable
Proposed System-draw the detailed block diagram to highlight the major modules in your system
Detail the algorithms/techniques(cite the algorithms/ models adopted from research papers here) you will be using for implementing each block in your proposed system
Datasets used
Add any other relevant details

4.1 Design and Methodology 1

4.2 Design and Methodology 2

4.3 Design and Methodology 3

...
Use page breaks after each section

5 Results and Discussion

Note: Use subsections and subsubsections wherever required throughout the report
Write the details of implementation/results obtained/relevant screenshots/any relevant discussion and comparative analysis of methodologies used Label all tables and figures wherever required

5.1 Results and Discussion 1

5.2 Results and Discussion 2

...
Add any other relevant details
Use page breaks after each section

6 Conclusion and Further Work

Note: Use subsections and subsubsections wherever required throughout the report summarise your conclusions and further work to be done for the next semester

7 Research Paper

Note: Use subsections and subsubsections wherever required throughout the report include the paper with plagiarism report here, if accepted for publication, attach the acceptance receipt/certificate if any

8 Plagiarism Report

Put your plagiarism report here
Use page breaks after each section

References

[1] Y. Srinivasa Rao and M. C. Chandorkar 'Numerical Integration Methods for Digital Power Electronics and Electrical Drives', *Proceedings of the National Conference on Computational Intelligence for Electrical Engineering*, Sant Longowal Institute of Engg. and Tech., Punjab, India, pp. 20-24, 18-19 Nov. 2005.

[2] S. Ben Saoud, B. Dagues, H. Schneider, M. Metz and J.C. Magiot, 'Real Time Emulator of Static Converters/Electrical Machines Application to the Test of Control Unit,' *Proceedings of the IEEE Symposium on Industrial Electronics*, vol. 2, pp. 856-861, June 1996.

[3] Joep Jacobs, Dirk Detjen, Claus-Ulrich Karipidis, and Rik W. De Doncker, 'Rapid prototyping tools for power electronic systems: Demonstration with shunt active power filters,' *IEEE Transactions on Power Electronics*, vol. 19, no. 2, pp. 500-507, March 2004.

[4] Antonello Monti, Enrico Santi, Roger A. Dougal, and Marco Riva, 'Rapid prototyping of digital controls for power electronics,' *IEEE Transactions on Power Electronics*, vol. 18, no. 3, pp. 915-923, May 2003.

[5] Y. Srinivasa Rao and Mukul Chandorkar, 'Rapid Prototyping Tool for Electrical Load Emulation using Power Electronic Converters', under review for *Proceedings of the IEEE Symposium on Industrial Electronics and Applications*, Kuala Lumpur, Malaysia, 4-6 Oct. 2009.

[6] Y. Srinivasa Rao and M. C. Chandorkar, 'Load Emulation with Power Electronic Converters', *Proceedings of the National Power Electronic Conference*, Indian Institute of Science, Bangalore, India, 17-19 Dec. 2007.

[7] V. Kaura, and V. Blasko 'Operation of a phase locked loop system under distorted utility conditions,' *IEEE Transactions on Industrial Applications*, vol. 33, no. 1, pp. 58-63, Feb. 1997.

[8] Marian P. Kazmierkowski and Luigi Malesani, 'Current control techniques for three-phase voltage-source PWM converters: A survey,' *IEEE Transactions on Industry Applications*, vol. 45, no. 5, pp. 691-702, Oct. 1998.

[9] Y. Srinivasa Rao and Mukul Chandorkar, 'Electrical Load Emulation using Power Electronic Converters', *Proceedings of the IEEE Region 10 Conference*, Hyderabad, India, 19-21 Nov. 2008, ISBN:978-1-4244-2408-5.

[10] Shin-ichi Hamasaki Hamasaki and Atsuo Kawamura, 'Improvement of current regulation of line-current-detection-type active filter based on deadbeat control,' *IEEE Transactions on Industry Applications*, vol. 39, no. 2, pp. 536-540, April 2003.

[11] Marco Liserre, Frede Blaabjerg, and Steffan Hansen, 'Design and Control of an LCL-Filter-Based Three-Phase Active Rectifier,' *IEEE Transactions on Industry Applications*, vol. 41, no. 5, pp. 1281-1291, Sept. 2005.

[12] Y. Srinivasa Rao and Mukul Chandorkar, 'Electrical Load Emulation using Optimal Feedback Control Technique', *Proceedings of the IEEE International Conference on Industrial Technology*, Gippsland, Australia, 9-13 Feb. 2009.

[13] Y. Srinivasa Rao and Mukul Chandorkar, 'Real-Time Electrical Load Emulator using Optimal Feedback Control Technique', under review for *IEEE Transactions on Industrial Electronics*.

[14] R. R. Sawant and M. C. Chandorkar, 'A Multifunctional Four-leg Grid Connected Compensator,' *IEEE Transactions on Industry Applications*, vol. 45, no. 01, pp. 249-259, Jan/Feb 2009.